

An Introduction to General Relativity: Spacetime and Geometry Solution Manual

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Chapter 1

Special Relativity and Flat Spacetime

1.1 Problem 13

(a) We have $F^{0i} = E^i$ and $F^{ij} = \epsilon^{ijk} B_k$

$$\begin{aligned}\mathcal{L}' &= \epsilon_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma} \\ &= \epsilon_{0\nu\rho\sigma} F^{0\nu} F^{\rho\sigma} + \epsilon_{1\nu\rho\sigma} F^{1\nu} F^{\rho\sigma} + \epsilon_{2\nu\rho\sigma} F^{2\nu} F^{\rho\sigma} + \epsilon_{3\nu\rho\sigma} F^{3\nu} F^{\rho\sigma}\end{aligned}\quad (1.1)$$

Consider the first term in the equation

$$\begin{aligned}\epsilon_{0\nu\rho\sigma} F^{0\nu} F^{\rho\sigma} &= \epsilon_{01\rho\sigma} F^{01} F^{\rho\sigma} + \epsilon_{02\rho\sigma} F^{02} F^{\rho\sigma} + \epsilon_{03\rho\sigma} F^{03} F^{\rho\sigma} \\ &= 2 (\epsilon_{0123} F^{01} F^{23} + \epsilon_{0231} F^{02} F^{31} + \epsilon_{0321} F^{03} F^{21}) \\ &= 2 (E^1 \cdot B^1 + E^2 \cdot B^2 - E^3 \cdot (-B^3)) \\ \epsilon_{0\nu\rho\sigma} F^{0\nu} F^{\rho\sigma} &= \mathbf{E} \cdot \mathbf{B}\end{aligned}\quad (1.2)$$

Similarly we can work out the other terms giving

$$\epsilon_{1\nu\rho\sigma} F^{1\nu} F^{\rho\sigma} = 2 (E^3 B^3 - E^2 B^2 + E^1 B^1) \quad (1.3)$$

$$\epsilon_{2\nu\rho\sigma} F^{2\nu} F^{\rho\sigma} = 2 (E^3 B^3 + E^2 B^2 - E^1 B^1) \quad (1.4)$$

$$\epsilon_{3\nu\rho\sigma} F^{3\nu} F^{\rho\sigma} = 2 (-E^3 B^3 + E^2 B^2 + E^1 B^1) \quad (1.5)$$

Adding these four equations we get

$$\mathcal{L}' = 4\mathbf{E} \cdot \mathbf{B} \quad (1.6)$$

Chapter 2

Manifolds

2.1 Problem 6: Tangent curves along given path

We have,

$$x = \cos \lambda \quad y = \sin \lambda \quad z = \lambda \quad (2.1)$$

From equations it is clear that the path is a Helix. Now, transforming to spherical polar coordinates,

$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{1 + \lambda^2} \quad (2.2)$$

$$\theta = \tan^{-1} \left(\frac{\sqrt{x^2 + y^2}}{z} \right) = \tan^{-1} \left(\frac{1}{\lambda} \right) \quad (2.3)$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right) = \tan^{-1} (\tan \lambda) \quad (2.4)$$

Chapter 3

Curvature

3.1 Problem 1

$$\nabla_\sigma \delta_\beta^\alpha = 0 \quad (3.1)$$

$$\nabla_\sigma (g^{\alpha\mu} g_{\mu\beta}) = 0 \quad (3.2)$$

$$(\nabla_\sigma g^{\alpha\mu}) g_{\mu\beta} + g^{\alpha\mu} (\nabla_\sigma g_{\mu\beta}) = 0 \quad (3.3)$$

$$(\nabla_\sigma g^{\alpha\mu}) g_{\mu\beta} = 0 \quad (3.4)$$

$$\nabla_\sigma g^{\alpha\mu} = 0 \quad (3.5)$$

where in (3.3) we have used product rule of covariant derivative.

3.2 Problem 2

This follows from Weinberg¹

- **Gradient:** Gradient is directly covariant derivative of a Scalar, i.e.

$$(\vec{\nabla}\phi)_\lambda = \nabla_\lambda \phi \quad (3.6)$$

Hence, in spherical polar coordinates

3.3 Problem 3

$$\Gamma_{\mu\nu}^\lambda = \frac{g^{\lambda\alpha}}{2} (\partial_\mu g_{\alpha\nu} + \partial_\nu g_{\mu\alpha} - \partial_\alpha g_{\mu\nu}) \quad (3.7)$$

- When all μ, ν and λ are different in (3.7) the christoffel symbol to be non-vanishing must have $\alpha = \lambda$ but then metric terms inside the parenthesis vanish and hence

$$\Gamma_{\mu\nu}^\lambda = 0 \quad (3.8)$$

¹S. Weinberg, Gravitation and Cosmology: Principles and Applications of the General Theory of Relativity, Wiley, 1972, Sec. 4.7, p. 106.

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$$\begin{aligned}
\Gamma_{\mu\mu}^{\lambda} &= \frac{g^{\lambda\alpha}}{2} (\partial_{\mu} g_{\alpha\mu} + \partial_{\mu} g_{\mu\alpha} - \partial_{\alpha} g_{\mu\mu}) \\
&= -\frac{g^{\lambda\lambda}}{2} (\partial_{\lambda} g_{\mu\mu}) \\
&= -\frac{(g_{\lambda\lambda})^{-1}}{2} (\partial_{\lambda} g_{\mu\mu})
\end{aligned} \tag{3.9}$$

When in last step we have used the fact that when the metric is diagonal $g_{\mu\nu} = (g^{\mu\nu})^{-1}$.

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$$\begin{aligned}
\Gamma_{\mu\lambda}^{\lambda} &= \frac{g^{\lambda\alpha}}{2} (\partial_{\mu} g_{\alpha\lambda} + \partial_{\lambda} g_{\mu\alpha} - \partial_{\alpha} g_{\mu\lambda}) \\
&= \frac{g^{\lambda\lambda}}{2} (\partial_{\mu} g_{\lambda\lambda} + \partial_{\lambda} g_{\mu\lambda} - \partial_{\lambda} g_{\mu\lambda}) \\
&= \frac{g^{\lambda\lambda}}{2} (\partial_{\mu} g_{\lambda\lambda}) \\
&= \frac{(g_{\lambda\lambda})^{-1}}{2} (\partial_{\mu} g_{\lambda\lambda}) \\
&= \partial_{\mu} \ln \sqrt{|g_{\lambda\lambda}|}
\end{aligned} \tag{3.10}$$

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$$\begin{aligned}
\Gamma_{\lambda\lambda}^{\lambda} &= \frac{g^{\lambda\alpha}}{2} (\partial_{\lambda} g_{\alpha\lambda} + \partial_{\lambda} g_{\lambda\alpha} - \partial_{\alpha} g_{\lambda\lambda}) \\
&= \frac{g^{\lambda\lambda}}{2} (\partial_{\lambda} g_{\lambda\lambda} + \partial_{\lambda} g_{\lambda\lambda} - \partial_{\lambda} g_{\lambda\lambda}) \\
&= \frac{(g_{\lambda\lambda})^{-1}}{2} (\partial_{\lambda} g_{\lambda\lambda}) \\
&= \partial_{\lambda} \ln \sqrt{|g_{\lambda\lambda}|}
\end{aligned} \tag{3.11}$$

3.4 Problem 4

Given transformation is

$$x = uv \cos \phi \quad y = uv \sin \phi \quad z = \frac{1}{2} (u^2 - v^2)$$

(a) The transformation matrix is trivial to find using $\partial x^{\alpha} / \partial x'^{\beta}$

$$\begin{pmatrix} v \cos \phi & u \cos \phi & -uv \sin \phi \\ v \sin \phi & u \sin \phi & uv \cos \phi \\ u & -v & 0 \end{pmatrix} \tag{3.12}$$

(b) The inverse transformation can be found out easily

$$\begin{aligned}
\tan \phi &= \frac{y}{x} \\
\phi &= \tan^{-1} \left(\frac{y}{x} \right)
\end{aligned} \tag{3.13}$$

For other need some small calculation

$$x^2 + y^2 = u^2 v^2 \quad (3.14)$$

Use above relation in the transformation of z i.e.

$$\begin{aligned} u^2 &= 2z + v^2 \\ u^2 &= 2z + \frac{x^2 + y^2}{u^2} \\ 0 &= u^4 - 2zu^2 - (x^2 + y^2) \\ u^2 &= \frac{2z \pm \sqrt{4z^2 + 4(x^2 + y^2)}}{2} \end{aligned} \quad (3.15)$$

3.5 Problem 5

(a) The geodesic equation is

$$\frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\rho\sigma}^\mu \frac{dx^\rho}{d\lambda} \frac{dx^\sigma}{d\lambda} = 0 \quad (3.16)$$

We now need non vanishing Christoffel symbols

$$\Gamma_{\theta\phi}^\phi = \cot \theta \quad \Gamma_{\phi\phi}^\theta = -\sin \theta \cos \theta \quad (3.17)$$

Now we have geodesic Equations

$$\frac{d^2 \theta}{d\lambda^2} - \sin \theta \cos \theta \left(\frac{d\phi}{d\lambda} \right)^2 = 0 \quad (3.18)$$

$$\frac{d^2 \phi}{d\lambda^2} + 2 \cot \theta \frac{d\theta}{d\lambda} \frac{d\phi}{d\lambda} = 0 \quad (3.19)$$

Now for lines of constant longitude i.e. $\phi = \text{const}$ will satisfy (3.19) and hence are geodesic.

Now for lines of constant latitude to satisfy (3.18) must have $\theta = \pi/2$, and hence will be geodesic equation.

(b) Choose the affine parameter λ to be ϕ . We have initial vector $V^\mu = (1, 0)$. Use of equation for parallel transport

$$\frac{dV^\mu}{d\phi} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{d\phi} V^\rho = 0 \quad (3.20)$$

$$\frac{dV^\theta}{d\phi} - \sin \theta \cos \theta V^\phi = 0 \quad (3.21)$$

$$\frac{dV^\phi}{d\phi} + \cot \theta V^\theta = 0 \quad (3.22)$$

differentiate (3.21) w.r.t. ϕ and use the result in (3.22)

$$\frac{d^2 V^\theta}{d\phi^2} + \cos^2 \theta V^\theta = 0 \quad (3.23)$$

putting $\omega = \cos \theta$, gives the solution

$$V^\theta = A \cos \omega \phi + B \sin \omega \phi \quad (3.24)$$

Suppose we start from $\phi = 0$ then using initial conditions We have $A = 1$. using (3.21) we have

$$V^\phi = \frac{1}{\sin \theta} (-\sin \omega \phi + B \cos \omega \phi) \quad (3.25)$$

Again using initial conditions $B = 0$ Hence final solution is

$$(V^\theta, V^\phi) = \left(\cos(\cos \theta \times \phi), -\frac{\sin(\cos \theta \times \phi)}{\sin \theta} \right) \quad (3.26)$$

After one round trip the components are

$$(V^\theta, V^\phi) = \left(\cos(2\pi \cos \theta), -\frac{\sin(2\pi \cos \theta)}{\sin \theta} \right) \quad (3.27)$$

Note: that for equator $\theta = \pi/2$ which is geodesic the final components are $(1, 0)$, agreeing with result that vector is unchanged when parallel transported round trip around geodesic.

3.6 Problem 6

(a)

Since clock is stationary $dr = d\phi = d\theta = 0$ therefore

$$\tau_1 = \sqrt{1 - \frac{2GM}{R_1}} t \approx t \left(1 - \frac{GM}{R_1} \right) \quad (3.28)$$

$$\tau_2 \approx t \left(1 - \frac{GM}{R_2} \right) \quad (3.29)$$

since $R_2 > R_1 \implies \tau_2 > \tau_1$; clock on higher altitude runs faster

(b)

3.7 Problem 8

(a) We'll find the Christoffel symbols by varying the integral

$$S = \int \left\{ \left(\frac{d\psi}{d\lambda} \right)^2 + \sin^2 \psi \left[\left(\frac{d\theta}{d\lambda} \right)^2 + \sin^2 \theta \left(\frac{d\phi}{d\lambda} \right)^2 \right] \right\} d\lambda \quad (3.30)$$

Varying with respect to ψ gives

$$\int \left\{ 2 \frac{d\psi}{d\lambda} \frac{d\delta\psi}{d\lambda} + 2 \sin \psi \cos \psi \left[\left(\frac{d\theta}{d\lambda} \right)^2 + \sin^2 \theta \left(\frac{d\phi}{d\lambda} \right)^2 \right] \delta\psi \right\} d\lambda = 0 \quad (3.31)$$

use integration by parts now

$$\int \left\{ \frac{d^2\psi}{d\lambda^2} - \sin \psi \cos \psi \left[\left(\frac{d\theta}{d\lambda} \right)^2 + \sin^2 \theta \left(\frac{d\phi}{d\lambda} \right)^2 \right] \right\} \delta\psi d\lambda = 0 \quad (3.32)$$

Therefore,

$$\Gamma_{\theta\theta}^\psi = -\sin \psi \cos \psi \quad \Gamma_{\phi\phi}^\psi = -\sin \psi \cos \psi \sin^2 \theta \quad (3.33)$$

Similarly varying with other coordinates gives us other christoffel symbols. The non vanishing ones are

$$\begin{aligned} \Gamma_{\theta\theta}^\psi &= -\sin \psi \cos \psi & \Gamma_{\phi\phi}^\psi &= -\sin \psi \cos \psi \sin^2 \theta \\ \Gamma_{\phi\phi}^\theta &= -\sin \theta \cos \theta & \Gamma_{\theta\psi}^\theta &= \cot \psi \\ \Gamma_{\psi\phi}^\phi &= \cot \psi & \Gamma_{\theta\phi}^\phi &= \cot \theta \end{aligned} \quad (3.34)$$

(b) Non vanishing components of Riemann tensor are:

$$\begin{aligned} R_{\theta\psi\theta}^\psi &= \partial_\psi \Gamma_{\theta\theta}^\psi - \partial_\theta \Gamma_{\theta\psi}^\psi + \Gamma_{\psi\eta}^\psi \Gamma_{\theta\theta}^\eta - \Gamma_{\theta\eta}^\psi \Gamma_{\theta\psi}^\eta \\ &= -\cos^2 \psi + \sin^2 \psi + \sin \psi \cos \psi \cot \psi \\ &= \sin^2 \psi \end{aligned} \quad (3.35)$$

$$R_{\phi\psi\phi}^\psi = \sin^2 \theta \sin^2 \psi \quad (3.36)$$

$$\begin{aligned} R_{\psi\theta\psi}^\theta &= \partial_\theta \Gamma_{\psi\psi}^\theta - \partial_\psi \Gamma_{\theta\psi}^\theta + \Gamma_{\theta\eta}^\theta \Gamma_{\psi\psi}^\eta - \Gamma_{\psi\eta}^\theta \Gamma_{\theta\psi}^\eta \\ &= \operatorname{cosec}^2 \psi - \cot^2 \psi \\ &= 1 \end{aligned} \quad (3.37)$$

$$\begin{aligned} R_{\phi\theta\phi}^\theta &= \partial_\theta \Gamma_{\phi\phi}^\theta - \partial_\phi \Gamma_{\theta\phi}^\theta + \Gamma_{\theta\eta}^\theta \Gamma_{\phi\phi}^\eta - \Gamma_{\phi\eta}^\theta \Gamma_{\theta\phi}^\eta \\ &= -\cos^2 \theta + \sin^2 \theta - \cos^2 \psi \sin^2 \theta + \cos^2 \theta \\ &= \sin^2 \theta \sin^2 \psi \end{aligned} \quad (3.38)$$

$$\begin{aligned} R_{\psi\phi\psi}^\phi &= \partial_\phi \Gamma_{\psi\psi}^\phi - \partial_\psi \Gamma_{\phi\psi}^\phi + \Gamma_{\phi\eta}^\phi \Gamma_{\psi\psi}^\eta - \Gamma_{\psi\eta}^\phi \Gamma_{\phi\psi}^\eta \\ &= \operatorname{cosec}^2 \psi - \cot^2 \psi \\ &= 1 \end{aligned} \quad (3.39)$$

$$\begin{aligned}
R_{\theta\phi\theta}^{\phi} &= \partial_{\phi}\Gamma_{\theta\theta}^{\phi} - \partial_{\theta}\Gamma_{\phi\theta}^{\phi} + \Gamma_{\phi\eta}^{\phi}\Gamma_{\theta\theta}^{\eta} - \Gamma_{\theta\eta}^{\phi}\Gamma_{\phi\theta}^{\eta} \\
&= \operatorname{cosec}^2\theta - \cot^2\theta - \sin\psi\cos\psi\cot\psi \\
&= 1 - \cos^2\psi \\
&= \sin^2\psi
\end{aligned} \tag{3.40}$$

The metric components are $g_{\psi\psi} = 1$, $g_{\theta\theta} = \sin^2\theta$ and $g_{\phi\phi} = \sin^2\psi\sin^2\theta$
Ricci Tensor

$$R_{\mu\nu} = R_{\mu\lambda\nu}^{\lambda} \tag{3.41}$$

$$R_{\psi\psi} = 2 \tag{3.42}$$

$$R_{\theta\theta} = 2\sin^2\psi \tag{3.43}$$

$$R_{\phi\phi} = 2\sin^2\theta\sin^2\psi \tag{3.44}$$

The metric components are $g_{\psi\psi} = 1$, $g_{\theta\theta} = \sin^2\psi$ and $g_{\phi\phi} = \sin^2\psi\sin^2\theta$
Ricci Scalar is

$$\begin{aligned}
R &= g^{\mu\nu}R_{\mu\nu} \\
&= 6
\end{aligned} \tag{3.45}$$

(c)

$$R_{\mu\nu\rho\sigma} = g_{\mu\eta}R_{\nu\rho\sigma}^{\eta} \tag{3.46}$$

Therefore,

$$\begin{aligned}
R_{\psi\theta\psi\theta} &= \sin^2\psi \\
R_{\psi\phi\psi\phi} &= \sin^2\theta\sin^2\psi \\
R_{\theta\phi\theta\phi} &= \sin^2\theta\sin^4\psi
\end{aligned} \tag{3.47}$$

Others follow from property of Riemann Tensor. Now to verify

$$\begin{aligned}
R_{\mu\nu\rho\sigma} &= \frac{R}{n(n-1)}(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}) \\
&= g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}
\end{aligned} \tag{3.48}$$

as $n = 3$

$$\begin{aligned}
R_{\psi\theta\psi\theta} &= g_{\psi\psi}g_{\theta\theta} - g_{\psi\theta}g_{\theta\psi} \\
&= \sin^2\psi
\end{aligned} \tag{3.49}$$

$$\begin{aligned}
R_{\psi\phi\psi\phi} &= g_{\psi\psi}g_{\phi\phi} - g_{\psi\phi}g_{\phi\psi} \\
&= \sin^2\psi\sin^2\theta
\end{aligned} \tag{3.50}$$

$$\begin{aligned}
R_{\theta\phi\theta\phi} &= g_{\theta\theta}g_{\phi\phi} - g_{\theta\phi}g_{\phi\theta} \\
&= \sin^4\psi\sin^2\theta
\end{aligned} \tag{3.51}$$